

4	(a) <u>add small mass</u> to cause extension then remove mass to see if spring returns to original length	M1	
	repeat for larger masses and note maximum mass for which, when load is removed, the spring does return to original length	A1	[2]
	(b) Hooke's law requires force proportional to extension graph shows a straight line, hence obeys Hooke's law	B1 M1	[2]
	(c) $k = \text{force} / \text{extension}$ $= (0.42 \times 9.81) / [(30 - 21.2) \times 10^{-2}]$ $= 47 \text{ (46.8) N m}^{-1}$	C1 C1 A1	[3]